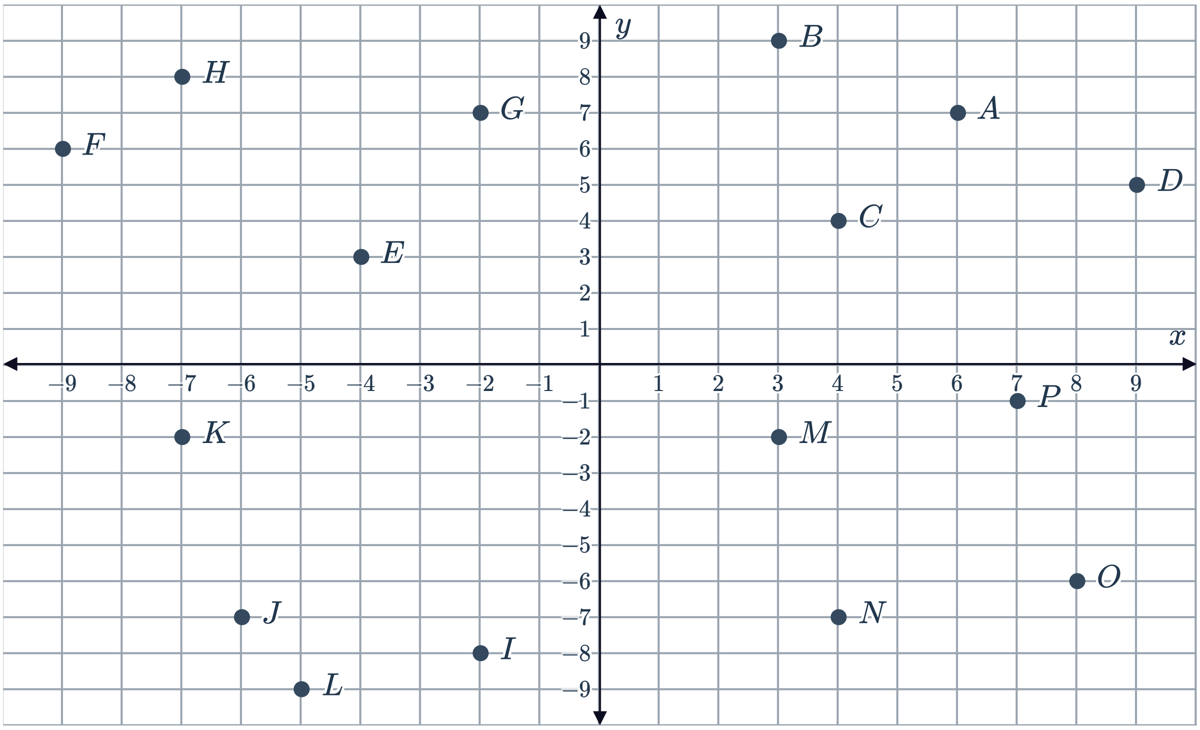
Directed numbers on the number pl

1 Consider the following number plane:



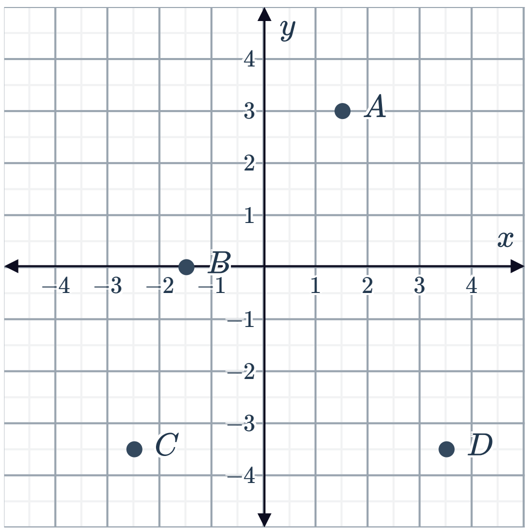
State the coordinates of the following points:

a A	**b** B	**c** C	**d** D
e E	**f** F	**g** G	**h** H
i I	**j** J	**k** K	**l** L
m M	**n** N	**o** O	**p** P

2 Consider the following number plane:

State the coordinates of the following points:

a A **b** B **c** C **d** D



3

a State the x-coordinate of:

i The origin.

ii Any point on the y-axis.

b State the y-coordinate of:

i The origin.

ii Any point on the x-axis.

4 If point P has an x-coordinate of 0, which axis must it lie on?

5 If (a, b) is a point that lies on the y-axis, find the value of a.

6 Consider the given points on the number plane:

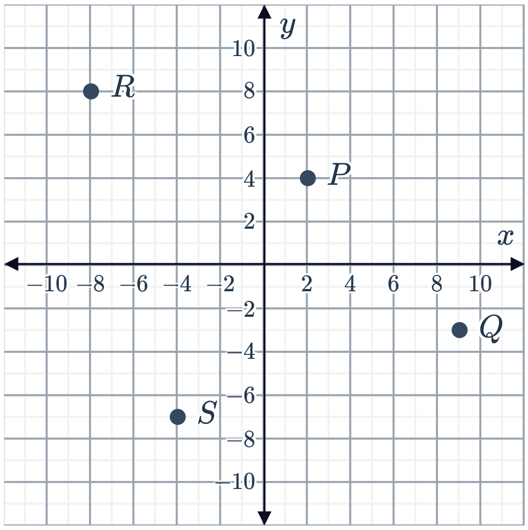
State which point has the following coordinates:

a $(9, -3)$

b $(-4, -7)$

c $(2, 4)$

d $(-8, 8)$

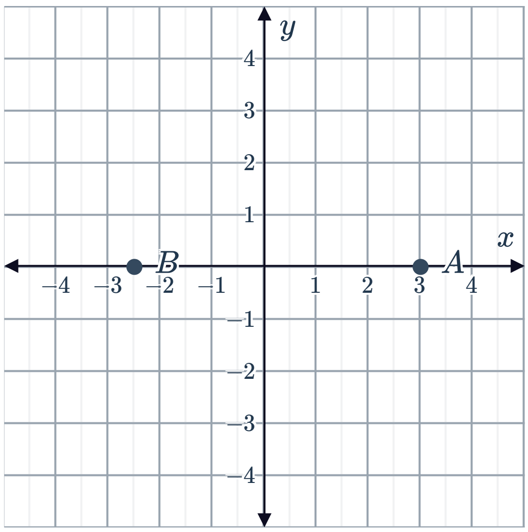


7 Consider the points A and B plotted on the number plane:

a Write down the coordinates of point A.

b Write down the coordinates of point B.

c Which axis do points A and B lie on?



8 In which quadrant do the following points lie?

a $(7, 8)$

b $(-3, 4)$

c $(2, -1)$

d $(-5, -6)$

9 State the quadrant(s) that have:

- **a** Points with a negative y-coordinate.
- **b** Points with a negative x-coordinate.
- **c** Points where the x-coordinate and y-coordinate have the same sign.
- **d** Points with a negative x-coordinate and a positive y-coordinate.
- **e** Points with a negative x-coordinate and a negative y-coordinate.

10 Plot the following points on a number plane:

a $A\,(6,1)$	**b** $B\,(-7,3)$	**c** $C\,(7,-4)$	**d** $D\,(-7,-3)$
e $E\,(-4,-4)$	**f** $F\,(-6,0)$	**g** $G\,(2,6)$	**h** $H\,(2,-8)$
i $I\,(-6,3)$	**j** $J\,(-5,-5)$	**k** $K\,(-5,9)$	**l** $L\,(5,-9)$
m $M\,(3,-3)$	**n** $N\,(0,8)$	**o** $O\,(1,3)$	**p** $P\,(-4,5)$
q $Q\,(-2,-4)$	**r** $R\,(4,-3)$	**s** $S\,(1,0)$	**t** $T\,(0,2)$

11 For each of the following points:

- **i** Plot the point on a number plane.
- **ii** State which quadrant the point lies in.

a $(2,5)$	**b** $(-2,1)$	**c** $(-3,-5)$	**d** $(8,-5)$
e $(-8,-1)$	**f** $(4,-7)$	**g** $(0,2)$	**h** $(5,0)$

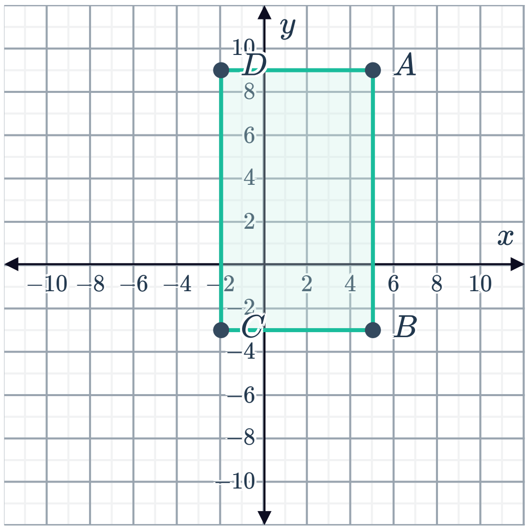
12 For each of the following sets of points:

- **i** Plot the points on a number plane.
- **ii** Identify the shape formed by joining the points in order.

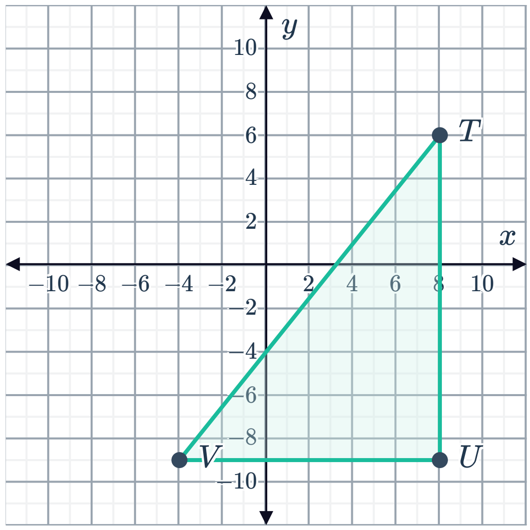
a $(4,1),(7,1),(4,4),(7,4)$	**b** $(1,5),(6,5),(1,8),(6,8)$
c $(-5,-3),(0,-1),(-4,6),(1,8)$	**d** $(-4,-3),(0,-1),(-2,1),(2,3)$

13 For each of the following shapes, write do[redacted]e coordinates of the vertices:

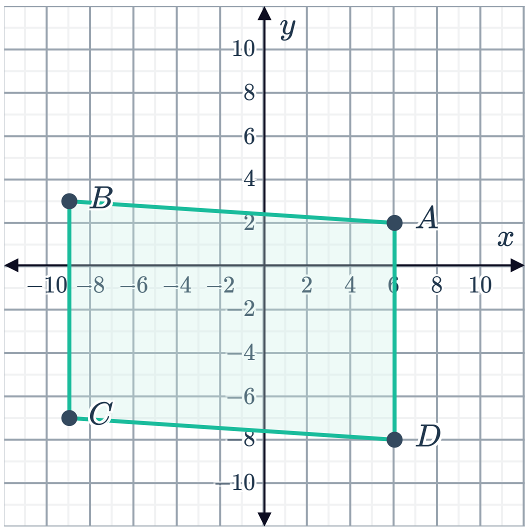
a



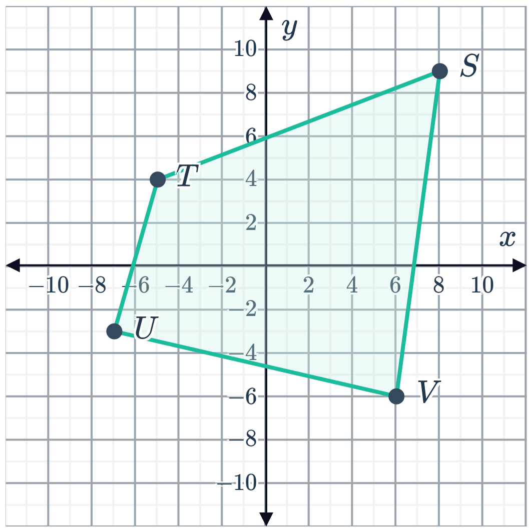
b



c



d



14 Describe how you would move on the number plane from the origin, to plot the following points:

a $(2, -4)$

b $(9, 14)$

15 Find the coordinates of the following points given the description:

a The point 9 units below the origin.

b The point 3.5 units to the left of the origin.

c The point 4 units to the left of $(-3, 6)$.

d The point 7 units to the right of $(-1, -2)$.

e The point 2 units to the right and 2 units below the point $(2, 5)$.

f The point 6 units to the left and 5 units above the point $(4, -4)$.

16 Which point is furthest from the origin?

A $(0,-3)$ **B** $(5,0)$ **C** $(0,4.5)$ **D** $(-4,0)$

17 How many units above the origin is the point $(-6,1)$ located?

18 Starting at the origin, move 4 units left and then 4 units up.

a. Plot the resulting point on a number plane.

b. State the coordinates of this point.

Distance between two points

19 Find the distance between the following pairs of points:

a. $A\,(-5,8)$ and $B\,(-2,8)$.

b. $A\,(7,3)$ and $B\,(-1,3)$.

c. $A\,(6,-5)$ and $B\,(6,-1)$.

d. $A\,(-6,2)$ and $B\,(-6,-7)$.

20 Consider the points: $A\,(-4,8)$, $B\,(-7,8)$ and $C\,(-7,1)$.

a. Plot the points on a number plane.

b. Find the length of AB.

c. Find the length of BC.

21 A triangle has points $A\,(1,2)$, $B\,(-2,-3)$ and $C\,(6,-3)$.

a. Plot the triangle ABC on a number plane.

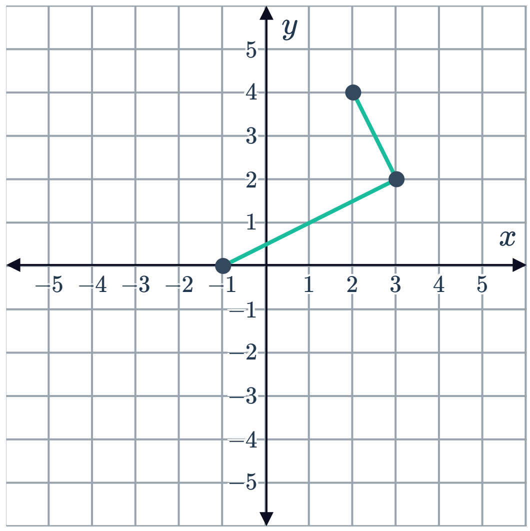
b. Find the perpendicular height of the triangle if BC is the base.

c. Find the length of base BC.

d. Find the area of triangle ABC.

Geometric applications

22 The points given represent three vertices of a parallelogram. Find the coordinates of the fourth vertex if it is known to be in the 2nd quadrant.



23 The points given represent three vertices of a rhombus. Find the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.